While exotic in detail, the broad-brush lesson illustrated by Figure 12.2 is one with which we are already familiar: concepts and conclusions relevant on one scale may not be applicable on all scales. This is a key principle in physics, and one that we encounter repeatedly even in far more prosaic contexts. Take a glass of water. Describing the water as a smooth, uniform liquid is both useful and relevant on everyday scales, but it's an approximation that breaks down if we analyze the water with sub-microscopic precision. On tiny scales, the smooth image gives way to a completely different framework of widely separated molecules and atoms. Similarly, Figure 12.2 shows that Einstein's conception of a smooth, gently curving, geometrical space and time, although powerful and accurate for describing the universe on large scales, breaks down if we analyze the universe at extremely short distance and time scales. Physicists believe that, as with water, the smooth portrayal of space and time is an approximation that gives way to another, more fundamental framework when considered on ultramicroscopic scales. What that framework is—what constitutes the "molecules" and "atoms" of space and time—is a question currently being pursued with great vigor. It has yet to be resolved.

Even so, what is thoroughly clear from Figure 12.2 is that on tiny scales the smooth character of space and time envisioned by general relativity locks horns with the frantic, jittery character of quantum mechanics. The core principle of Einstein's general relativity, that space and time form a gently curving geometrical shape, runs smack into the core principle of quantum mechanics, the uncertainty principle, which implies a wild, tumultuous, turbulent environment on the tiniest of scales. The violent clash between the central ideas of general relativity and quantum mechanics has made meshing the two theories one of the most difficult challenges physicists have encountered during the last eighty years.

Does It Matter?

In practice, the incompatibility between general relativity and quantum mechanics rears its head in a very specific way. If you use the combined equations of general relativity and quantum mechanics, they almost always yield one answer: infinity. And that's a problem. It's nonsense.